# Ankit Raj

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## Professional Summary

- Backend Software Engineer with 2+ years of experience building production systems using Java, Python, and PostgreSQL, with experience in microservices, distributed systems, and AI-powered applications.
- Strong open-source contribution, owner of one of the most popular libraries in Python called PyWhatKit having 2.5+ million users world-wide, contributed to some other open-source projects as well.

## Education

**B.Tech, Computer Science and Engineering** Sept 2020 – Sept 2024
*Dr. A.P.J. Abdul Kalam Technical University, Ghaziabad*

**Summer School on Digital Platform** Aug 2023
*Aalto University, Helsinki, Finland*

## Experience

**Software Engineer** Mar 2026 – Present
*Nextuple India Pvt. Ltd.* Bengaluru, India

– Built a Discrepancy Correction API for Supply and Demand services that automatically identified and fixed inventory mismatches, improving data consistency to **99.95%** while completing reconciliation in **under 10 minutes**.
*Tech: Java, Python, PostgreSQL, REST APIs, Microservices*

– Developed and maintained backend services for inventory and supply chain workflows, collaborating with cross-functional teams to deliver reliable, production-ready features.

**Software Developer** July 2024 – Feb 2026
*Aveti Learning Pvt. Ltd.* Bhubaneswar, India

– Core member of the engineering team, contributing to system design decisions and mentoring junior developers.

– Engineered and maintained a secure, production-grade backend for a US-based client **Dignifi**, enabling formerly incarcerated individuals to access housing, education, and job opportunities.
*Tech: FastAPI, PostgreSQL, JWT Auth, Docker, AWS*

– Designed and developed an Info-Search backend system leveraging vector embeddings to enable semantic search across 100K+ educational content items.
*Tech: FastAPI, PostgreSQL, pgvector, Sentence Transformers, Python*

– Led system design and backend development for a semantic image search application using CLIPViT, indexing 140K+ images and supporting similarity-based retrieval (continued from internship).
*Tech: Python, FastAPI, CLIPViT, pgvector, PostgreSQL*

– Developed an AI-based attendance system capable of training on a single facial image per student and accurately marking attendance from 3–4 group images of up to 60 students, achieving 94% accuracy and significantly reducing teacher effort.
*Tech: Python, OpenCV, Face Recognition, Deep Learning Models*

– Designed and implemented an AI-agent–based evaluation system that automatically grades multiple students' answer sheets with high accuracy using provided answer keys as training data.
*Tech: Python, LLMs, OCR, Prompt Engineering*

– Actively involved in backend architecture decisions, database schema design, performance optimization, and deployment workflows for AI-driven education products.

**Software Development Intern (Long-Term)** Jan 2021 – June 2024
*Aveti Learning Pvt. Ltd.* Remote

– Selected for this software development internship in the first year of college based on open-source contributions and creating Pywhatkit.

– Built the **Smart Class App** used by 1400+ schools across Odisha for offline smart board content delivery in low-connectivity environments.
*Tech: Python, PyQt, SQLite*

– Developed a PPT Translation and review backend used by 1200+ teachers for educational content validation and approval.
*Tech: Flask, MySQL, REST APIs*

– Indexed and managed 140K+ media files using vector embeddings to support a scalable Document Management System.
*Tech: PostgreSQL, pgvector, Python, Embedding Models*

– Created an offline e-library system deployed on Raspberry Pi devices, benefiting 600+ rural students with limited internet access.
*Tech: Raspberry Pi, Linux, Python*

## SKILLS

- **Languages**: Java, Python, SQL
- **Frameworks & Libraries**: FastAPI, Flask, Django, NodeJS
- **Databases & Search**: PostgreSQL, MySQL, MongoDB, Redis, Qdrant, Pgvector
- **Messaging & Streaming**: Kafka, RabbitMQ
- **DevOps & Tools**: Docker, Git, GitHub, AWS, Heroku, Linux, Kubernetes, Raspberry-Pi
- **AI & ML**: Vector Search, Embeddings, RAG, LLMs

## PROJECTS

### PyWhatKit GitHub

– Creator and maintainer of PyWhatKit, a beginner-friendly Python automation library with 2.5M+ users worldwide, offering features such as WhatsApp scheduling, YouTube search, ASCII art generation, and text-to-handwriting conversion.
*Tech: Python, Requests, Flask, OpenCV*

### Deep-Live-Cam GitHub

– Contributed a macOS build fix resolving installation failures on Intel-based systems for a **70K-star** open-source deepfake project, improving cross-platform compatibility and developer adoption.
*Tech: Python*

### Other Projects GitHub

– **HomeworkMachine**: Built a tool to convert typed text into realistic handwritten notes, used by **10K+ users**.
*Tech: Python, OpenCV, Image Processing*

– **PhotoCluster**: Implemented a photo clustering system for mobile images using vector embeddings and unsupervised learning.
*Tech: Python, KMeans, Qdrant, Vector Embeddings*

– **Clipry**: Developed an Electron-based clipboard manager for Linux with fast search and history management.
*Tech: Electron, JavaScript, Linux*

## ACHIEVEMENTS & CERTIFICATIONS

- Won **1st Prize** in Hackathons at ADGITM IPU and DTU during college.
- Selected for a **Finland Summer School** trip by Aveti Learning as a reward for hardwork.
- Got a **Job Offer** in the current organization for developing PyWhatKit
- **AI Agents Fundamentals** – HuggingFace
- **Redis AI, Vectors, Streams** – Redis University